

Foundation (Jalapeno)

- Q1.** Solve the system: $x^2 + y^2 = 4$, $x + y = 2$
(A) $(x, y) = (0, 2)$
(B) $(x, y) = (2, 0)$
(C) $x = 1, y = 1$
(D) $x = -1, y = -1$
(E) No solution
- Q2.** What is the graphical representation of $x^2 + y^2 = 4$?
(A) Line
(B) Circle
(C) Parabola
(D) Ellipse
(E) Hyperbola
- Q3.** Solve the system: $y = x^2 - 2$, $y = -x^2 + 2$
(A) $x = 0, y = -2$
(B) $x = 1, y = -1$
(C) $x = 0, y = 2$
(D) $x = -1, y = -1$
(E) Infinite solutions
- Q4.** What is the solution to the system: $x^2 - y^2 = 0$, $x + y = 2$?
(A) $x = 1, y = 1$
(B) $x = -1, y = -1$
(C) $x = 0, y = 2$
(D) $x = 2, y = 0$
(E) $x = -2, y = 0$
- Q5.** The system $x^2 + y^2 = 1$, $x - y = 1$ has
(A) One solution
(B) Two solutions
(C) No solution
(D) Infinite solutions
(E) Cannot be determined
- Q6.** The system $y = x^2$, $y = -x^2 + 4$ has
(A) Two solutions
(B) One solution
(C) No solution
(D) Infinite solutions
(E) Cannot be determined
- Q7.** Solve the system: $x^2 - 4y^2 = 1$, $x + 2y = 1$
(A) $x = 1, y = 0$
(B) $x = -1, y = 0$
(C) $x = 0, y = 1$
(D) $x = 0, y = -1$
(E) $x = 2, y = 1$
- Q8.** What is the graphical representation of $y = x^2 - 4$ and $y = x^2 + 2$?
(A) Two intersecting lines
(B) Two parallel lines
(C) Two intersecting parabolas
(D) Two parallel parabolas
(E) A circle and a parabola

Open Ended Questions (FRQ)

- Q9.** Solve the system $x^2 + y^2 = 4$, $x - y = 2$. Show all work and explain your answer.
- Q10.** Graph the system $y = x^2 - 1$, $y = -x^2 + 3$. Identify any points of intersection.

Chef's Tips

- Read each question carefully
- Use a pencil to write your answers
- Check your work